# LAB EXPERIMENT RECORD

## SET 18: Product Inventory Management

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|---|---|
| **Department** | AI&DS |
| **Course Code / Course Name** | DSA0602 / Data Handling and Visualization |
| **Experiment** | Data Visualization using R |
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| **Register Number** | 192524136 |
| **Date** | 10 OCTOBER 2026 |
| **Day** | Saturday |

**SET 18: Product Inventory Management**

**Aim:**

To visualize product inventory quantities and prices using R programs and an interactive dashboard in Tableau**.**

**Dataset: Product Inventory Management**

| Product ID | Product Name | Quantity Available | Price ($) |
|---|---|---|---|
| **1** | **A** | **250** | **20** |
| **2** | **B** | **175** | **15** |
| **3** | **C** | **300** | **18** |

**PROGRAM 1: Bar Chart (Product Quantity)**

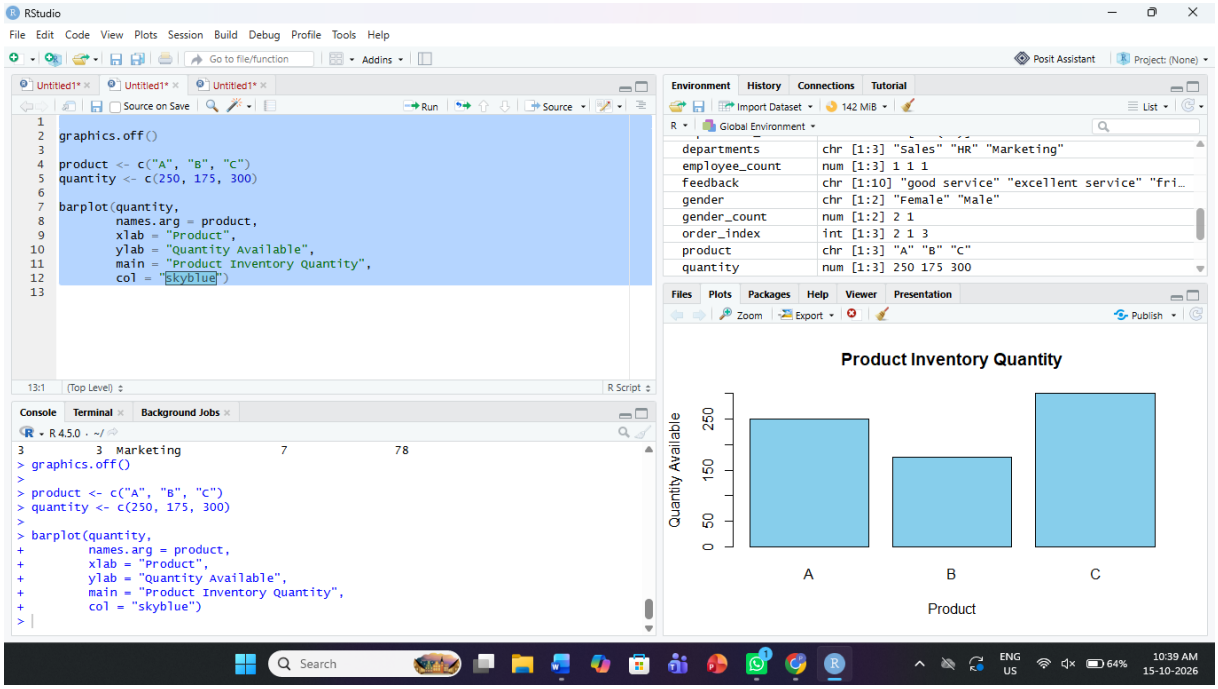
**Aim:** To display the available quantity of each product using a bar chart.

**R Program:**

```
graphics.off()

product <- c("A", "B", "C")
quantity <- c(250, 175, 300)

barplot(quantity,
    names.arg = product,
    xlab = "Product",
    ylab = "Quantity Available",
    main = "Product Inventory Quantity",
    col = "skyblue")
```




**Observation:** The bar chart compares the available quantities of three products. Product C has the highest quantity of 300 units, while Product B has the lowest quantity of 175 units.

**PROGRAM 2: Stacked Bar Chart (Product Inventory)**

**Aim:** To demonstrate a stacked bar chart for comparing product quantities across categories.

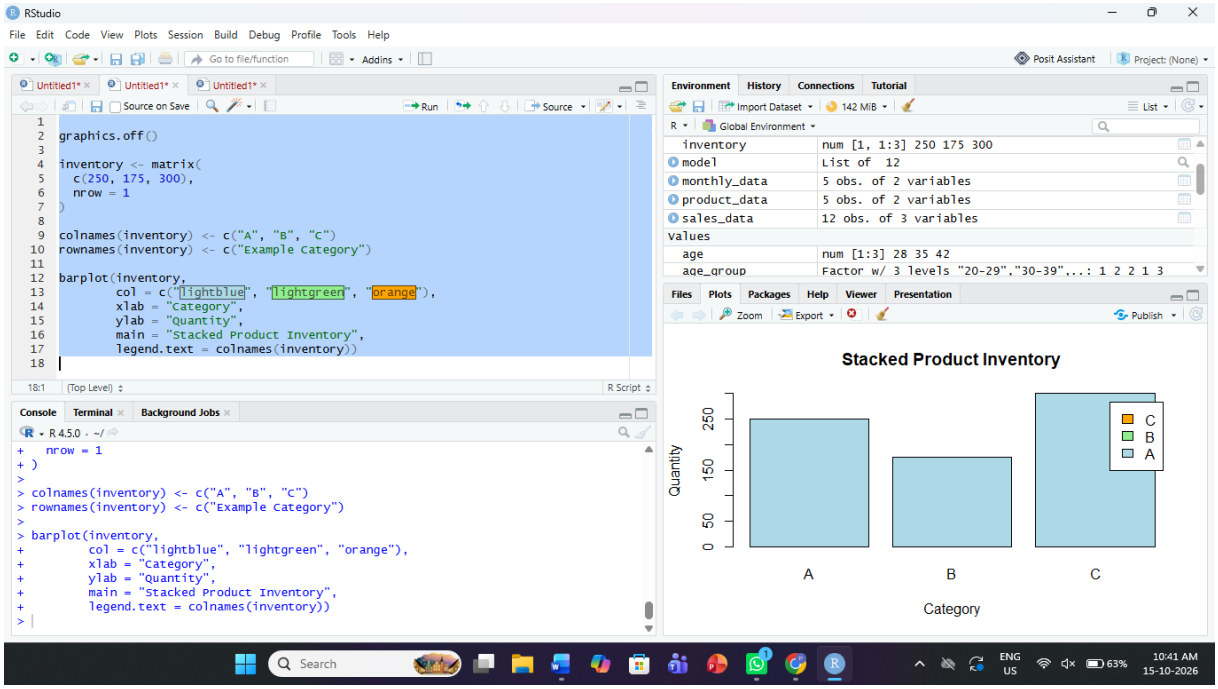
**R Program:**

```
graphics.off()

inventory <- matrix(
  c(250, 175, 300),
  nrow = 1
)

colnames(inventory) <- c("A", "B", "C")
rownames(inventory) <- c("Example Category")

barplot(inventory,
        col = c("lightblue", "lightgreen", "orange"),
        xlab = "Category",
        ylab = "Quantity",
        main = "Stacked Product Inventory",
        legend.text = colnames(inventory))
```




**Observation:** The stacked bar chart displays the quantities of Products A, B, and C within the example category. Actual category-wise comparisons require category information in the dataset.

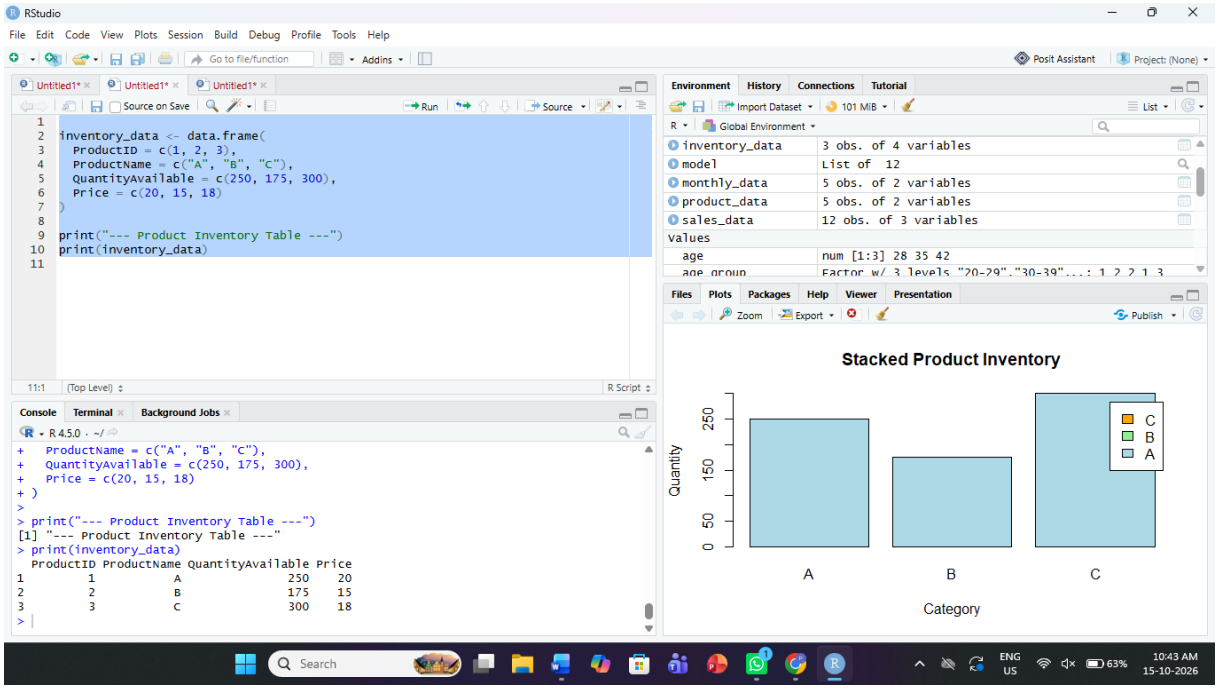
**PROGRAM 3: Product Inventory Summary Table**

**Aim:** To create a structured table summarizing product IDs, names, available quantities, and prices.

**R Program:**

```
inventory_data <- data.frame(
  ProductID = c(1, 2, 3),
  ProductName = c("A", "B", "C"),
  QuantityAvailable = c(250, 175, 300),
  Price = c(20, 15, 18)
)

print("--- Product Inventory Table ---")
print(inventory_data)
```



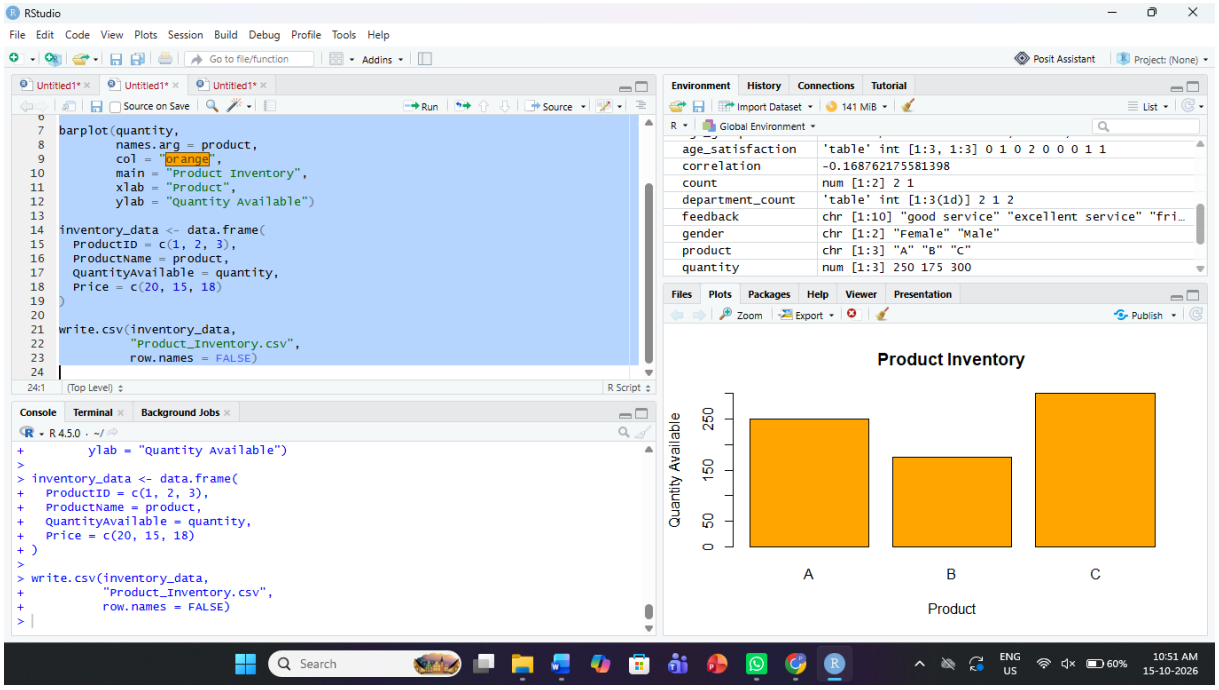

**Observation:** The table displays the product inventory details. Product C has the highest available quantity, while Product A has the highest price of $20.

**PROGRAM 4: Product Inventory Dashboard in Tableau**

**Aim:** To create an interactive dashboard combining product quantities, inventory comparisons, and product records**.**

**Steps in Tableau:**

1. Save the dataset as Product_Inventory.xlsx and connect it to Tableau.
2. Sheet 1 – Quantity Bar Chart: Drag Product Name to Columns and Quantity Available to Rows. Select Bar Chart.
3. Sheet 2 – Inventory Comparison: Display product quantities using a bar chart. To create a meaningful category-based stacked chart, first add the actual product category field to the dataset.
4. Sheet 3 – Inventory Data Table: Display Product ID, Product Name, Quantity Available, and Price.
5. Create a new dashboard and arrange the sheets.
6. Enable Use as Filter to filter the inventory information**.**




**Observation:** The dashboard combines product quantities and inventory records in one view. Filtering helps users examine individual products. Category-based analysis requires category data.

**Procedure:**

1. Create vectors containing product names and available quantities.
2. Use barplot() to display product quantities.
3. Use a matrix and barplot() to demonstrate stacked inventory visualization.
4. Create a data frame to display product inventory records.
5. Combine the visualizations in a Tableau dashboard.

**Result:**

Product inventory data was visualized using bar charts and a summary table in R. A Tableau dashboard was designed to support inventory analysis.